Jeonghan **Kim**

kimjh9813@hanyang.ac.kr | Kim-JeongHan | Kim-JeongHan

Research Interests

My research focuses on generative motion planning for high-dimensional robotic systems under complex constraints. I am particularly interested in diffusion and drifting models that learn multimodal trajectory distributions from offline data, with the goal of enabling efficient, feasible, and long-horizon planning.

Education

Hanyang University(ERICA) *Ansan, South Korea*

B.S. in Robot Engineering, minor in Electronic Engineering *Mar. 2018 - Feb. 2024*

- GPA: 3.91 / 4.5; Major GPA: 4.13 / 4.5

Research Experience

University of Michigan *Ann Arbor, MI, USA*

Visiting Student Researcher (Advised by Prof. Kang G. Shin) *Jul. 2026 - Aug. 2026*

- Research on generative path planning and offline decision-making, with a focus on single-pass trajectory generation and long-horizon planning.

RAISE Lab (Prof. Youngmoon Lee) *Hanyang University*

Undergraduate Researcher *Jan. 2023 - Feb. 2024*

- Develop manipulator-based HRI systems and a separable drone–mobile robot system.

Professional Experience

Roboe Technologies *Seoul, South Korea*

Software Engineer, Motion Algorithm Team *Mar. 2024 - Jun. 2026*

- Develop a reachability-based workspace analysis tool and pose optimization for conveyor-mounted mobile robots.
- Implement conveyor jamming detection and optimize PointCloud2 filtering to operate at 20 Hz.
- Design and implement a Behavior Tree-based motion-task framework for modular pick-and-place execution.
- Deploy a production pick-and-place system with throughput of 250–300 boxes per hour.
- Implement C++ motion planning algorithms for moveL and moveJ trajectories in virtual controllers.

Gole Robotics *Seoul, South Korea*

Intern, Autonomous Driving Software *Aug. 2023 - Oct. 2023*

- Implement a packet-based serial communication library aligned with embedded controller protocols.
- Benchmark IMU, LiDAR, ToF, and barometer candidates using mean/variance analysis and PlotJuggler to support sensor selection.

Publications

Snapbot: Enabling Dynamic Human–Robot Interactions for Real-Time Computational Photography *ACM/IEEE HRI 2024* *Boulder, USA*

C. Choi, J. Kim, Y. Nam, and Y. Lee

- We propose a perception-to-control pipeline for dynamic camera composition, combining head-pose tracking with pseudo-inverse Jacobian velocity control.

Separable Integrated Drone–Mobile Robot System for Unmanned Patrol *ICROS 2023* *Samcheok, South Korea*

J. Kim, J. Park, H. Lee, S. Park, Y. Nam, G. Kwon, W. Jung, and Y. Lee

- We propose a separable drone–mobile robot system with integrated indoor/outdoor autonomous navigation for unmanned patrol.
- ICROS 2023 — **Undergraduate Best Paper Award**

Selected Projects

Planning - Python 3D Path Planning Library *GitHub (Open Source)*

Creator & Main Contributor *Sep. 2025 - Present*

- Implement sampling-based path-planning algorithms: RRT, RRT-Connect, RRG, RRT*, Informed RRT*, PRM, and PRM*.
- Implement Euclidean and terrain-aware Riemannian metrics for N-dimensional path planning.
- Implement Diffuser for generative trajectory planning.

Snapbot: Computational Photography and HRI Robot *Hanyang University*

Manipulator Control & Perception *Sep. 2023 - Dec. 2023*

- Develop a manipulator-based computational photography robot that tracks people and selects camera poses for dynamic human-robot interaction.
- Design a YOLO/depth-based human perception pipeline to estimate the relative TF between the robot and a detected person.
- Implement inverse-kinematics closed-loop manipulator control with filtering and damped pseudo-inverse singularity avoidance.
- IITP President Award — **2nd Place Nationwide**, 2023 SW Talent Festival National Capstone Competition.

Separable Drone-Mobile Robot System for Unmanned Patrol *Hanyang University*

Project Lead *Jan. 2023 - Jun. 2023*

- Lead development of an integrated drone-mobile robot system where a ground robot carries, charges, and deploys a drone for last-mile patrol over rugged terrain.
- Design ROS architecture, indoor/outdoor autonomous operation, app-to-ROS interface, and point-cloud based terrain perception modules.
- Fuse IMU, GNSS, and wheel odometry with Kalman filtering to reduce localization error to 0.5%.

ERIC_A: Indoor Autonomous Delivery Robot *Hanyang University*

Project Lead *Sep. 2022 - Dec. 2022*

- Lead a 6-member indoor delivery robot project integrating web order dispatch, ROS navigation, and YOLO-based human detection.
- Improve odometry error from 5–10% to 1% through IMU/encoder Kalman-filter fusion.
- Achieve 30 mm $\pm$ 1 destination accuracy at 0.5 m/s average robot speed.

Hongdo: Public-Deployment HRI Mascot Robot *Ansan City, South Korea*

Project Lead *Jul. 2022 - Nov. 2022*

- Lead an 8-member team in developing a mobile mascot robot for deployment at public festivals.
- Integrate a ROS-based mobile robot architecture with LiDAR obstacle avoidance, human following, and a web-based HRI interface for AI photo transformation.
- Demonstrate the robot to 200–300 citizens and iterate on system reliability between festival deployments.
- Place 2nd in the Ansan Young Innovators Design Thinking Workshop and receive **KRW 20M** in project funding from Ansan City.

Awards

Nov. 2023	**IITP President Award — 2nd Place Nationwide**, National Capstone Competition — Snapbot	*South Korea*
Jun. 2023	**Top Award**, Intelligent Robot AI Idea Competition	*South Korea*
Aug. 2022	**2nd Place**, Ansan Young Innovators Design Thinking Workshop	*South Korea*
Jul. 2022	**1st Prize**, Cancer Patient Care and Health Product Idea Competition	*South Korea*
Aug. 2021	**1st Prize**, ODA Idea Competition	*South Korea*

Teaching & Service

IEEP Tanzania International Science Volunteer *Arusha, Tanzania*

Volunteer *Jan. 2024*

- Participate in an international science volunteer program.
- Contribute to science education.

HYMEC - Robot Making Club *Ansan, South Korea*

Club President *Jan. 2022 - Dec. 2022*

- Lead member activities for a robotics club with over 100 members.
- Organize mobile-robot mission hackathons and peer robotics activities for approximately 200 participants.

Tutoring Mentor *Ansan, South Korea*

- Mentor students in Git, ROS/ROS2, and Docker.

Technical Skills

Software	C/C++, Python, PyTorch, Ray, Git, Docker, pytest, mypy, ruff, uv
Robotics	ROS1/2, MoveIt2, motion planning, collision detection, PointCloud2, Open3D
Robot Platforms	UR e-Series, RB20, TM20, HDR20-17, mobile robots